AberdeenGroup

Competing for the IBM AS/400 Market

A Report to the Management of Sun Microsystems

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Introduction

Sun Microsystems asked Aberdeen Group to assess the attractiveness of IBM's AS/400 and iSeries market, and to recommend how Sun might profitably attack the iSeries market.

The iSeries is the current incarnation of proprietary AS/400 midrange server family launched in 1988. This server family has seen over 750,000 units sold in 14 years, and contributed untold billions in profits to IBM. We use the AS/400 and iSeries terms synonymously throughout the report.

This report is the result of Aberdeen's research project. The Aberdeen Group estimates, analysis, and recommendations contained in this report are confidential to Sun Microsystems, and may not be made public without the written permission of Aberdeen Group, Inc.

Executive Summary

Demographics and Market Conditions

- The AS/400 installed base is 450,000 machines, of which approximately 25,000 are running the latest OS/400 release on currently offered hardware.
- We estimate that in 2001 the iSeries hardware generated $7.68 billion in revenue representing 23% of IBM's hardware revenues, and $2.59 billion in software representing 20% of 2001 software revenues. This makes the iSeries a $10.27 billion business for IBM. We estimate a gross margin contribution of $7.2 billion — and a COGS of 30%.
- Revenues have declined approximately 25% since the pre-Y2K peak in 1998 at $13.5 billion. Erosion is caused by deliberate server consolidation, competition at price-sensitive accounts by Windows NT, and replacement by more modern applications (e.g., Siebel and SAP). However, the revenue decline should not be taken as a sign that the iSeries base is ripe for picking. Growth is coming in emerging markets (China), mergers and acquisitions, an expanding partner-driven distribution system, deeper penetration into the SMB market, and
- IBM believes the iSeries has an addressable market of $56 billion (Table 3). Based on our estimates, IBM is capturing 18.3% of this addressable market.
- iSeries is focused at specific vertical markets through business partners who complete 85%+ of transactions. Key verticals which represent 55% of sales are manufacturing, finance, and whole/retail (distribution) (Table 4).

- Besides a vertical focus, IBM is currently focusing on new-application upgrades with business partners: CRM, ERP, SCM, and B2B.
- IBM sells the iSeries globally. We estimates 2001 hardware and software revenues in North America at $3.96 billion, EMEA at $4.0 billion, and Asia/Pacific at $2.3 billion.
- The typical customer has a one- or two-processor system (value $50K-$150K), representative of a small-medium business (SMB) with fewer than 1,000 employees. To that SMB customer, the iSeries is their "mainframe".

Competitive Assessment

- The iSeries has five models scaling 500-fold from roughly 400 TPM-C (model 250) to 375,000 (model 890), with base prices starting under $10,000 (more like $15,000 configured) (Table 9). Full complement of peripherals. Native support for IBM's eBusiness infrastructure. Good ability to manage Intel-based servers. OS/400 is a mature commercial operating system with an embedded (object-)relational database. No obvious gaps that Sun could exploit with the exception that IBM in general is not offering Linux-based appliances.
- The iSeries benefits from one of the industry's leading business partner programs, empowering partners around the globe with incentives and marketing. The ISV partner program over fifteen years has resulted in thousands of specialized applications. Partners are also a key component of the service and support infrastructure, typically providing first-level support.
- There is a mature and broad set of application development tools from IBM and partners for the development of modern eBusiness applications using Web infrastructure as well as legacy, dumb-terminal applications.
- Price-performance is relatively poor as the iSeries is premium-priced, particularly on upgrades. However, customers are generally satisfied with TCO since the iSeries is easy to operate and administer (e.g., no Oracle DBA required).
- Europe is the key battle ground. Any iSeries program of Sun must have a large investment in understanding and attracting IBM's targeted European SMB market.
- To the SMB buyers who want IBM to provide solutions, the iSeries positioning and messages are well crafted, tested, and working (Table 8).

IBM's 2002 iSeries Game Plan

- Server consolidation: replace old AS/400s and centralize.
- Upgrade e-infrastructure: IGS-driven WebSphere.
- Competitive Target: HP3000
- SMB new-market growth: Notes Domino, portals, and mid-market ERP with Navision business partner.

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- Enterprise Application Solutions: modernize customer applications in CRM/BI, ERP, and SCM.
- Selectively target small business market (under 100 employees).
- Replace or refinance Y2K-era machines.

AS/400 Displacement Market Opportunity

Factors Positive to Sun Microsystems

1. *Technology Transition*: The iSeries is beginning a technology transition that will soon result in common hardware between IBM's Unix pSeries (e.g., p670 and p690 "Regatta") and Integrated Application iSeries. The positive aspects of this (see also the negative aspects below) are the increased competition between IBM business partners, caused by the lack of hardware differentiation between pSer ies (AIX Unix) and iSeries.
2. *ISV Competition:* By marrying AIX and OS/400 in one chassis (see #1 above), IBM will unwittingly unleash two ISV camps that until this point were largely mutually exclusive; relatively few customers — and almost no SME customers — have both IBM Unix *and* AS/400 ISV applications.
3. *RAS*: The transition to the Regatta micro-architecture means iSeries customers must buy into IBM's new keep-running-or-fail philosophy — when a hard failure occurs, the Mean Time to Repair will be quite long compared to Sun due to the complexity of the backplane and multi-chip carrier design.
4. *Linux*: Many iSeries shops made a "no Unix" decision up to a decade ago and stuck with it. But the rapid emergence of Linux in the enterprise, heartily supported by IBM, creates a new opportunity for Sun.
5. *Harmonious Trends*: Server consolidation is a marketing thrust of IBM, coinciding with an IT re-centralization trend, and continued interest in eBusiness application development leading to web services.
6. Sun has superior price and price-performance.

Factors Negative to Sun Microsystems

1. *More than 90% of the iSeries and AS/400 installed base is a combination of trailing edge, rather-die-than-switch mentality*. These operational managers grew up with the AS/400, have had ten years to evaluate Unix (and by implication Sun), and are not subject to a rational business discussion about the right platform supplier for their enterprise's future.
2. *IBM has done a good job of keeping the iSeries current in hardware and software* — successfully transitioning to RISC in the late 1990s and by offering partitioning, Lotus Domino, native Linux and Java, and management of xSeries servers. These technologies have allowed IBM's AS/400 to hold off Unix for a decade.

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Soon, IBM will announce that AS/400 customers can have it *both* ways, with a Regatta-class iSeries server able to run both OS/400 and AIX in different partitions. *We conclude that a technology-based market attack on the AS/400 would be relatively easy for IBM to deflect*.

3. *The AS/400 market is not static*. IBM allows district sales and country offices to experiment with new marketing and customer support programs as buyer interests and economic conditions change.
4. *IBM Global Services is a major recommender of ISV-based solutions sales.* They work with IBM business partners, and wouldn't think of recommending Sun.
5. About *85%-90% of iSeries sales are made by IBM business partners* located in some 150 countries. IBM treats most ISVs as VARs, creating an economic incentive for the business partner to win with IBM. The iSeries is truly a global business for IBM, and Sun will have to be very aggressive to match IBM's coverage.
6. *The lease renewal window which opened up for post-Y2K replacements is closing*. Three-year leases have already rolled. Five-year leases are up this year and next year.

Sun's Addressable Market and Opportunity Size

Table 13 depicts in tabular form Aberdeen's analysis of the market opportunity and sizing for Sun based on flat IBM revenues for the iSeries in 2002 at $10.27 billion. Our conclusion is that the addressable market for Sun in 2002 is $1.87B:

- New Customers — represent 10% of iSeries revenues, $1,027M. We believe Sun can address 90% of this market, yielding an addressable market of $924M.
- Consolidation — represents 6% of IBM iSeries revenues. We believe 65% is addressable by Sun, yielding an addressable market of $400M.
- Applications — CRM, EAS, SCM, and B2B represent 11% of iSeries revenues. We believe that 40% is addressable, yielding an addressable market of $452M.
- Upgrades — Hardware and software upgrades related to Linux and xSeries management are estimated at 3% of revenues. This is the only portion of iSeries upgrades we find attractive. But selling Sun platforms as upgrades into iSeries shops is a low yield proposition, so we estimate only 30% is addressable — yielding $92M in addressable market for Sun.

Installed Base Migration is an Unattractive Market

iSeries customers leaving the IBM installed base are not covered in the addressable market above. First, we do not observe an identifiable class of dissatisfied iSeries users, so we cannot define an addressable market. Second, iSeries attrition to other platforms is already addressed by Unix and Windows NT competitors, so those sales are already being counted. Third, 75% of the installed base is in a low-growth/no-growth state of inertia. These customers are happy with their AS/400s and see little need to upgrade, let alone switch to Sun.

Recommended Market Strategy for Sun Microsystems

Aberdeen Group's recommended market strategy is to target selected segments of the iSeries market, as identified in the Addressable Market Opportunity section above.

1. New Customers

IBM wins new customers through its worldwide business partner organization. Sun has been and is already competing for these sales. Follow IBM, and partner with a selected set of IBM's partners (e.g., Sun at J. D. Edwards).

2. Applications — CRM, EAS, SCM, B2B

IBM, Sun, and Aberdeen all believe there is an attractive market for enterprise applications encompassing customer relationship management, ERP and enterprise applications, supply chain management, and business-to-business Internet infrastructure. IBM's present target is the small-medium business segment of this market. Aberdeen believes Sun should counter this program with its own.

3. iSeries Upgrades — Linux is the Lever

In Table 13, we show the addressable market for upgrades as 3% of revenues, worth $308 million, whereas the total upgrades market is more than an order of magnitude greater. Sun cannot compete for the typical memory, CPU, or storage upgrades. Sun can compete for Linux and Linux appliance business, with a projected capture of $92 million. IBM supports Linux on the iSeries as a guest OS. But IBM has done a poor job of packaging Linux with appliance functionality. We believe the overall enterprise level of interest in Linux, and the number of Linux-based applications, will skyrocket. Sun's market opportunity is to offer Linux to iSeries customers as an alternative to IBM's OS upgrade.

4. Consolidation

Consolidation often means centralizing small, distributed AS/400 system workloads back to a central data center. Sun is well positioned for enterprise-class, mission-critical commercial computing. A key technology and services issue will be the ability to migrate existing applications off of the AS/400 and onto Solaris, but Sun has considerable experience in its successful mainframe migration program.

Market Partner Strategies

Sun has to fight IBM's business partner "fire" with like fire: any Sun program that is not partner-driven will either fail or not reach the potential market. The key to the iSeries castle is through business partners.

Systems Integrators
Sun already has business relationships with the major global systems integrators. Accenture, for example, has considerable experience with both iSeries and SPARC Solaris. Because the SIs are often called in to do the huge services portion of enterprise applications, they represent a short list for frank discussions. The SIs are a leverage point in the new-application market opportunity, where CRM, EAS, SCM, and B2B applications at the F2000 represent hundreds of millions in server revenues over the next several years.

ISVs and VARs
Any strategic move by Sun into the iSeries market would require a comprehensive comparative analysis of the IBM and Sun business partner programs, with a special emphasis on ISVs. IBM treats most ISVs like VARS, even if IBM gets the lead and does some of the selling, spreading enough money around to represent a serious portion of the ISV's marketing budget. Therefore, Sun needs to dissect IBM's business partner program in order to know the facts before it can develop a comprehensive counter program.

IBM's top-tier iSeries business partners are shown in Appendix B. A number of these (e.g., J.D. Edwards) are also members of iForce. Since these top business partners drive a large fraction of new iSeries sales, Sun should focus on these ISVs first. The smaller or more specialized ISVs are available by vertical market at IBM's business partner web site.

Detailed Market Overview

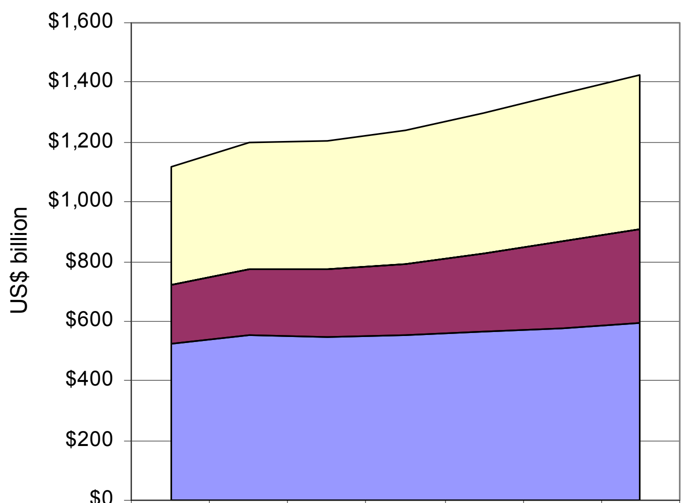
Worldwide IT Spending

The March 2002 report by Aberdeen Group, *World IT Spending 2002-2005: Timing the Recovery,* was used for sizing the world market and world geographies. This report is incorporated herein, and the report will be referred to as *WITS.*

Hardware, Software and Services

For 2001 to 2005, hardware spending growth will be eclipsed by expansion in software and services spending (Figure 1). Software sales will experience a CAGR of 8.1%, while services will increase at a CAGR of 4.8% and hardware will grow at a CAGR of 2.1%. Or, from another perspective, suppliers over the same timeframe are competing over $83.4 billion in additional software spending, $89.1 in new services revenues, and only $47.8 in additional hardware spending.

Figure 1 — Hardware, Software and Services Spending – World



	1999	2000	2001	2002	2003	2004	2005
Services	$394.6	$426.5	$428.6	$445.7	$470.1	$492.6	$517.7
Software	$197.4	$221.2	$228.9	$241.0	$261.4	$290.3	$312.3
Hardware	$525.3	$552.3	$545.1	$553.0	$567.2	$577.6	$592.9

Source: Aberdeen Group, March 2002, WITS

AS/400 and iSeries Installed Base

IBM has shipped 750,000 AS/400's and iSeries machines since June 1987, with 250,000 installed since 1998. IBM says that 450,000 machines are known to IBM, and are considered the "installed base" by IBM. Of those 450,000, about 25,000 use the latest re-

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lease of the OS/400 operating system, and thus could be considered "current generation". The machines are spread over 250,000 customers, or roughly two machines per customer.

The base is dramatically skewed, with the vast majority of units being 1- and 2-way machines, while the bulk of the revenue comes from the larger 8X0 models which are used by F2000 companies in up to 24-way configurations.

AS/400 and iSeries Market Size

We estimate that in 2001 the iSeries hardware generated $7.68 billion in revenue representing 23% of IBM's hardware revenues, and $2.59 billion in software representing 20% of 2001 software revenues. This makes the iSeries a $10.27 billion business for IBM. We estimate a gross margin contribution of $7.2 billion — and a COGS of 30%. This high gross margin fuels significant IBM marketing programs.

IBM-provided services add billions to this, plus value add by business partners. The entire iSeries ecosystem is over $20 billion — and roughly the size of Sun Microsystems.

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Market Segmentation

Business Size

Aberdeen has analyzed IBM's business size definitions, which define Small Business as 1-99 employees, Medium Business as 100-999 employees and Large Business as 1,000 or more employees. A historical analysis of IBM's fifteen year experience with the AS/400 and iSeries reveals that the iSeries installed base — and current sales — are not distributed as the total market by business size. The AS/400 market is strongly skewed towards the mid-market. IBM estimates of the growth by business size breakdown are shown in Table 2.

Table 2 — Segmentation by Business Size

Customer Set	CAGR 02-05 Intel Servers	CAGR 02-05 All Servers
Small Business (1-99 employees)	17.4%	9.5%
Medium Business (100-999 employees)	8.6%	6.8%
Large Business (1000+ employees)	12.8%	4.2%
Government	10.2%	4.2%

Source: IBM

The distribution of market opportunity according to IBM is distributed as shown in Table 3. IBM considers Table 3 the addressable market for the iSeries.

Table 3 — iSeries Addressable Market According to IBM

Customer Set	Market Opportunity 2002 $B
Small Business (1-99 employees)	$15.1
Medium Business (100-999 employees)	$09.5
Large Business (1000+ employees)	$19.4
Government	$12.1
Total	$56.1

Source: IBM Market Intelligence

Aberdeen Group's analysis of the demographics of U.S. businesses as provided by the federal government are shown in Appendix A. We do not have comparable information for Europe and Asia.

AS/400 Market Verticals

IBM is very focused on selling effectively into vertical markets. The iSeries is especially strong in wholesale/retail distribution, manufacturing, healthcare, and telco billing. The only area of relative weakness is in the high-tech vertical market.

IDC is apparently undercounting AS/400's, as they have only identified 226,000 servers in data provided Aberdeen Group by Sun, while IBM claims 450,000 active systems. The following table shows our estimates for the installed base of AS/400's and iSeries machines based on factoring IDC's numbers to the 450,000 base, and IBM and Aberdeen estimates for market share penetration by vertical market.

Table 4 — iSeries & AS/400 Vertical Market Distribution

IBM Vertical	WW Base	Percent of Base	Est. Market Share
Manufacturing	112,227	24.9%	16%
Financial	70,938	15.8%	12%
Retail/Wholesale	61,783	13.7%	16%
Health	45,837	10.2%	8%
Other	42,138	9.4%	9.5%
Professional Services	40,198	8.9%	9%
Telco/ISP	24,192	5.4%	4%
Computer/IT	23,949	5.3%	3%
Government	19,341	4.3%	5%
Education	9,398	2.1%	4%
Total	450,000	100.0%	

Source: Aberdeen Group, IBM, and IDC

Aberdeen Group's model for IT spending is broken down by the categories shown in Table 5. Our "Super Groups" consolidate specific industries as shown in the federal government's Bureau of Economic Analysis' industrial categorization. The lines from the BEA form are shown to explain how Aberdeen defines its vertical market model. Included are our estimates of each vertical's contribution percentage and estimated 2001 IT spending for hardware, software and services.

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Table 5 — Aberdeen Group Vertical Market Model

Vertical Market Super Group	Super Group %	Industry	Lines from US Bureau Economic Affairs Table	Adjusted IT Spending	2001 U.S. IT Spending in $Billions
Agriculture, Forestry, Fishing, Hunting and Mining	0.51%	Agriculture, Forestry, Fishing, Hunting and Mining	4, 8, 11	0.51%	$2.27
Construction	3.29%	Construction	12	3.29%	$14.73
Manufacturing	21.27%	Aerospace/ Defense	23 + Gov. Def. Estimate	3.84%	$17.21
		Automotive	22	1.14%	$5.09
		Consumer Packaged Goods	27, 28, 31	2.07%	$9.28
		Other Manufacturing	Lines 14 through 36, except 21, 22, 23, 33 and 35	11.25%	$50.36
		Pharmaceutical/Chemical	33, 35	2.97%	$13.30
Transportation and Utilities	5.90%	Energy	9, 10, 50	3.61%	$16.18
		Transportation and Warehousing	39, 40, 41, 42, 43, 46	2.29%	$10.23
Financial Services	17.06%	Banking	54	5.63%	$25.22
		Brokerage and Securities Services	55, 56, 62	7.54%	$33.75
		Insurance	57, 58	3.89%	$17.42
Business Services	10.08%	Other Business Services (non-	66, 72 Minus IT consulting	5.65%	$25.28

Vertical Market Super Group	Super Group %	Industry	Lines from US Bureau Economic Affairs Table	Adjusted IT Spending	2001 U.S. IT Spending in $Billions
		Financial)			
		Media/PR	49, 69	1.27%	$5.67
		Real Estate	59	3.17%	$14.18
Information Technology	10.92%	IT Services/ Consulting	WITS Calc	1.65%	$7.40
		Computer Hardware/Software	WITS, Includes part of line 21	4.52%	$20.24
		Telecommunications Equipment and Services	48 + $50 Other; Includes part of Line 21	4.75%	$21.25
Wholesale/ Retail	15.47%	Retail	52	7.40%	$33.13
		Air Travel and Hospitality	44, 64	1.70%	$7.60
		Wholesale Trade	51	6.38%	$28.55
Healthcare	5.17%	Healthcare	71	5.17%	$23.16
Government and Education	5.24%	Education	73	0.37%	$1.66
		Government	79 - part of Aerospace/ Defense	4.87%	$21.79
Other	5.08%	Other	65, 67, 68, 70, 74, 75, 76, 77	5.08%	$22.74
	100.00%	Total/Ave		100.00%	$447.7

Source: Aberdeen Group Vertical Market base research, WITS, U.S. BEA

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iSeries by Application

In addition to providing vertical market solutions through business partners for the verticals discussed above, IBM has a worldwide program underway to gain market share in CRM, SCM, B2B, and enterprise applications (integrated ERP). IBM believes the solutions opportunity approximates Table 6.

Table 6 — Application Solutions Opportunity

$B	2001	2002	2003	2004	2005	CAGR
CRM	34	42	52	63	75	22%
EAS	23	25	29	33	37	12%
SCM	15	20	26	33	35	23%
B2B	4	6	7	9	11	26%

Source: IBM

Aberdeen believes these CAGR estimates are inflated by pre-recession optimism. However, we do not dispute IBM's conclusion that these are the correct four new horizontal application areas. Our conclusion is that Sun should fight IBM for this new business at iSeries customers.

In addition to verticals, IBM is focusing on horizontal opportunities:

- business intelligence (as part of CRM)
- e-mail and collaboration, particularly with Lotus Domino
- Server consolidation, centralizing distributed systems and managing Intel-based xSeries systems running Windows NT
- Linux-on-iSeries as an alternative to adding more servers (with higher cost of administration) specifically for Linux applications.

iSeries by Geography

Our forecasts for worldwide and geographical breakdown of IT spending are contained in *WITS*. Aberdeen's key findings from that March 2002 report are as follows:

- *IT spending will begin to recover in 2002*. Worldwide IT spending in 2001 grew at a meager 0.2%, while declining 0.4% in the US. In 2002, Aberdeen forecasts that world IT spending will grow at 3.1% and US spending will increase at 4.3%.
- *The real IT spending recovery will begin in the second quarter of 2002*. Aberdeen expects US spending to improve 3.9% when measured against Q2 2001 and world IT spending to increase by 2.7%.

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- *But long-term growth rate will be less than that experienced in the late 1990's.* Aberdeen does not forecast a return to double-digit growth rates. Rather, we believe that for 2003 – 2005, world IT spending will increase at an annual rate of 4% to 5% and US spending will grow at 5% to 6%.
- *The factors contributing to excessive growth in the late 1990s are not repeatable*. These elements included corporate retooling around Y2K issues, the Internet/e-Commerce euphoria phenomenon, excessive venture capital investment in dot.coms, traditional enterprises accelerating e-business deployments to match the dot.com threat, and unsustainable IT spending in the telecom sector.
- *Long-term IT spending is gated by GDP growth and corporate revenue growth*. IT spending now accounts for 3.85% of world GDP and 4.65% of US GDP. Additionally, the US Department of Commerce reports that over 65% of US GDP consists of consumer spending, making IT an even greater percentage of business spending and investment. IT spending decisions now factor heavily into total business expenses and must be budgeted accordingly.
- *As a result, sub-segments exhibiting hyper-growth will do so at the expense of others*. IT spending is no longer a tide that will raise all boats. Supplier competition over limited budget dollars will get increasingly fierce.
- *China's importance will increase dramatically*. Aberdeen expects that China will vault from the seventh largest market for IT products and services (in 2001) to the fourth largest market by 2005, surpassing the UK, France and Italy.
- *Japan continues to languish*. A decade of low growth and capital investments has taken it toll. Japan lags other developed nations in current IT spending and will not return to positive growth levels until 2003.
- *The top 10 countries account for nearly 75% of world spending*. The U.S. alone accounts for 37.5%. Obviously, IT spending power is concentrated in a handful of markets.
- *The U.S. will lead the recovery*. Aberdeen expects that US IT spending growth will exceed the world average for 2002 – 2005.
- *Software and services spending gain at the expense of hardware*. Worldwide, hardware expenditures will increase a total of only 8.7% from 2001 to 2005, while software and services will increase 36% and 21% respectively over the same time period.

Aberdeen estimates iSeries hardware and software revenues for 2002 as depicted in Table 7. In general, the SMB market focus of IBM tends towards lower-end machines in EMEA and AsiaPac. However, large, national industries such as banking make the iSeries high end a commonly found machine in every major geography. Brazil and India have a relatively sparse AS/400s and iSeries installed base due to past government market limits on imports. China is attractive for growth rates and low technology skills for operation and administration, which IBM views as iSeries strengths.

Table 7 — iSeries Distribution by Model and Geography

Model ($Ms)	2001 Revenue	North America	EMEA	Asia Pacific
250	$250	$90	$95	$65
270	$3,500	$1,225	$1,330	$945
820	$2,500	$1,000	$975	$525
830	$2,018	$807	$807	$404
840	$2,000	$840	$800	$360
Total	$10,268	$3,962	$4,007	$2,299

Aberdeen Group estimates

Segment-Specific AS/400 Penetration

As shown in Table 4, Vertical Market Base, almost 55% of AS/400s are from the following industries:

1. Manufacturing — across the whole gamut of this "horizontal" vertical, IBM partners have applications for all aspects of process, discrete, batch, build-to-order, shop floor control, and specialty manufacturing. Inventory control is probably the most common use of the AS/400, along with financials.
2. Finance — industry-specific applications for retail and wholesale banking and insurance are most common. These applications exist in many countries where they have been tailored to local business and regulatory requirements.
3. Wholesale/Retail/Distribution — more inventory control, warehouse control, POS, and financials.

Many of the iSeries customers are SMBs, where the cost of a large IT staff is not affordable.

Overview of ISV AS/400 Offerings And Support

The iSeries offering is broad, and delivered in almost every country. Key attributes are:

- Five models scaling 500-fold from roughly 400 TPM-C to 200,000, with base prices starting under $10,000 (more like $15,000 configured).
- Full complement of peripherals.

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- Native support for IBM's eBusiness infrastructure.
- A mature commercial operating system with an embedded (object-)relational database.
- Logical Partitioning (LPAR), the industry's finest granularity in partitioning, is tested and mature.
- Native support for DB2, Lotus Domino, Linux, and support for directly managing farms of Intel/Microsoft servers.
- Participation in IBM's eLiza technology initiative for self-managing computing systems.
- Extensive IBM Global Services provided worldwide, including training, installation, sizing and capacity planning, best practices, and outsourcing.
- One of the industry's leading business partner programs, empowering partners around the globe with incentives and marketing. The ISV partner program over fifteen years has resulted in thousands of specialized applications. Partners are also a key component of the service and support infrastructure, typically providing first-level support.
- A mature and broad set of application development tools from IBM and partners for the development of modern eBusiness applications using Web infrastructure as well as legacy, dumb-terminal applications.

iSeries Market Trends

Aberdeen believes that overall AS/400 and iSeries revenues have declined approximately 25% since the high-water mark of 1998 when pre-Y2K purchases drove hardware and software revenues to $13.5 billion. This gradual decline is caused not by loss of customer confidence in the iSeries but by the relentless pressure of more performance for a lower price. Even the pricey iSeries is not immune to industry trends. Nevertheless, the profits from the iSeries are critical to IBM's overall results, so we believe IBM will continue to invest in iSeries technology and marketing. We estimate marketing programs in 2002 at the $750 million level, enough to keep business partners and IBM sales focused on churning the installed base.

The primary reasons for installed base erosion are:

- Server consolidation is removing thousands of distributed systems. This trend will continue and is an attractive opportunity for Sun.
- Windows NT has captured a small percentage of smaller sites due to either better price-performance and/or superior selling by the Microsoft business partner.
- Replacement of older applications by newer applications, often in conjunction with server consolidation, in the past led to some off-migration. For example, Bristol Meyers Squibb replaced 125 country-level servers running the

BPICS software from System Software Associates (since bankrupt) with three data centers running SAP R3 on Unix.

Growth of the base is primarily due to:

- Emerging markets, particularly in Asia (e.g., China) where rapid business growth drives bigger businesses and distributed sites at new offices and plants.
- Normal business changes such as mergers and acquisitions, and normal business growth. For example, Microsoft to this day produces its distributor invoices on an iSeries that has been upgraded several times over the past decade.
- New conquests by an expanding business partner family. Called the iNation, IBM is calling attention to the unique iSeries ecosystem of product, partners, and customers.
- The new conquests are often in the global SMB markets, where the breadth and depth of the iSeries business partners pays off on the margin.
- The IBM mantle of service and support continues to attract a percentage of new customers who are ready, willing, and able to pay extra for the privilege of having IBM service them.

Current and Anticipated Migration Patterns

Incoming Migration

The factors effecting incoming migration are discussed above. There is no discernable trend towards the iSeries and away from any other platform. The exception is “other proprietary platforms”. IBM has already targeted the HP 3000 installed base due to the similarities with the AS/400 base: many F1000 customers; running large and profitable systems; mission-critical computing; mature and sensitive to an IBM services-led proposition; unwilling to go to Microsoft. There are other occasional wins as aged Unisys, Wang, Data General, Siemens-Nixdorf, Groupe Bull, ICL, and Digital Equipment machines are replaced.

Outgoing Migration

There is no significant out-migration to other platforms. IBM co-opted the Windows NT platform by allowing the iSeries to directly attach to Intel/Windows servers, sharing storage, and allowing the iSeries console to manage the attached NT systems.

iSeries shops evaluated Unix over the last ten years and made their choice: there is nothing to gain in pitching Solaris per se to AS/400 shops. IBM now supports Linux as a native OS in its own partitions on every iSeries machine. In addition, we believe that AIX will become available on new POWER4-based iSeries machines within the next eighteen months.

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Market's Ability To Provide In House AS/400 Technical Support
The AS/400 base is a hodge-podge of different development technologies:

- The old-timers (and there are many MIS managers who will retire before their AS/400 is replaced with some different architecture) still use RPG and the powerful OS/400 Command Line Interpreter for ad hoc work. New RPG programmers are not being trained in colleges or technical schools, so this development pool is graying — or doing in-house training.
- COBOL is another major development language having fewer legacy problems on the iSeries than the AS/400, but which suffers the same no-new-programmers problem as RPG, but has a somewhat larger base.
- C has been supported for a decade, and is widely used by ISVs.
- Java and IBM's WebSphere are fully supported, as is Lotus Notes and Domino. There are extensive training programs since IBM wants to move its entire eServer platform base to these lock-in development tools.

We do not rank the ability to provide technical support as high on the list of potential Sun value propositions.

Overview Of Other Market Characteristics

Business Partners
Business Partners are inextricably tied to the AS/400 market. About 85% of sales are made through partners. In the past year, IBM has rather aggressively responded to partner complaints about sharing the pie and IBM's investments:

- Implemented $18M in investment-based incentives
- Increased incentives by 34%
- Gave revenue credit to partners for $225M in ibm.com revenue
- Implemented a new territory model where partners are given opportunity management and control
- Offered a technology refresh with a 63% discount on an i820 base system, plus $58,000 in free WebSphere, Linux, and Notes Domino software, and a free integrated xSeries server to push development.

Platform Transition
The impending announcement of iSeries servers based on the POWER4 microprocessor will extend the performance range of the iSeries product line, and signal IBM's continued investment in the iSeries. This is both symbolic and important to the installed base, as any indication of an iSeries "end of life" (i.e., what HP did to the HP 3000 base) would be an opportunity Sun could not afford to pass up. There is no iSeries end of life on the horizon. Sun should look at the p690 and p670 as the basis for comparable midrange and high-end future iSeries.

Competitive Assessment

IBM's AS/400 Strategy

2002 Strategic Game Plan for iSeries and AS/400
Our assessment of IBM's big thrusts this year:

Server Consolidation: Replacing older AS/400s, often distributed to line-of-business locations (e.g., retail stores) with data center large servers. A win in server consolidation: protects the customer; sells a new generation of technology; and, doubles the hardware/software sale through services.

Typically IBM proposes to move old AS/400-proprietary workloads to the new server first. Note that a Sun sales efforts will likely fail on the inability to run the existing workloads. Then, IBM will upsell on the large server's ability to support:

- Linux
- LPAR (logical partitioning) ability to segregate and manage workloads
- X86 applications with Windows NT. IBM is running CIO seminars aimed at proving the iSeries has lower TCO when managing attached Wintel servers.
- Lotus Mail and iNotes
- High Availability configurations (i.e., clusters, partitions)
- New-breed applications using IBM's web services (e.g., WebSphere).

e-Infrastructure: Notably, IBM's:

- e-outsourcing services
- WebSphere projects and client retraining
- Third-party application rebuilding products such as Lansa and eOne.

Competitive Target of Opportunity: IBM is targeting HP's 3000 installed base since HP has declared the 3000 obsolete. IBM believes they can knock HP 9000s out of accounts that are disgruntled about the HP 3000 decision (including many F1000 companies).

IBM also continues to invest in new ISV recruitment and in building the eServer brand (so that the proprietary AS/400 no longer stands out like a sore thumb).

SMB New-Market Growth: Notes Domino (including dedicated servers), WebSphere Portal, and Navision lead the charge. IBM believes there is a $19B iSeries server opportunity this year at companies with fewer than 100 employees, and a $12B opportunity with businesses that are 100-999 employees. The SMB market has been a key focus for IBM for some years, and IBM's marketing and partner programs towards the SMB market are mature.

New Focus on Small Businesses: Defined as up to 99 employees, IBM believes there are over 100 small businesses for every large one. Unix and iSeries represent a $8.1B opportunity in 2002. There are 7.4M small businesses in the U.S. and over 21M in EMEA (IBM research). IBM is focusing on two categories:

- Premium IT Solution Adopters represent 5% of the category but 37% of server spending. These are looking for best of breed solutions which enable business processes, provide high reliability and high performance, but are highly demanding of IT deployment and consulting support.
- Value-minded adopters represent 9% of the universe and one-third of server opportunity. They are characterized as seeking best of breed IT products and services, but value-pricing branded products and services is critical.
- The two categories above are also looking for Wintel servers, which the iSeries manages rather elegantly.

Enterprise Application Solutions: Key solutions providers are J.D. Edwards, IBS, Intentia, SAP, Siebel, Manhattan Associates, and other CRM/BI, ERP, and SCM ISVs. These are assessed at $6B for ERP, $5B for CRM and $3B for SCM.

Y2K Era Replacements: About 62% of the AS/400s purchased in anticipation of Y2K were financed with three and five year leases. These machines are late in their lease renewal periods, or recently rolled over. IBM Finance wins if the lease is renewed. IBM product groups win if the machine is replaced.

Unfortunately from a timing standpoint, a Sun attack campaign will find many machines recently upgraded or refinanced and thus largely untouchable.

More LPAR: The fine granularity of the iSeries logical partitioning will also be available on the Unix pSeries this fall. IBM intends to make a major differentiation campaign in comparison to other Unix and Wintel alternatives. IBM finds that its LPAR technology story is easy to make to both technologists and line managers. But note that only 20% of iSeries machines actually leave the factory with LPAR turned on.

Europe is the Battleground

Any battle for the iSeries market must be won in Europe. Key factors include:

- IT spending in Europe is about one-half that of the U.S., and can be used to offset difficult labor and regulatory polices.
- Europe has twice as many SMB companies — some 21,000 — as in the U.S., which are the prime target for IBM.
- European Community-driven consolidation across Europe is occurring, driving business away from traditional “national” brands like Olivetti and Groupe Bull and towards multi-national brands.
- While Unix revenue was flat in 2001, iSeries was up 5% — an incremental $250M — while partners saw 11% Year-over-Year growth to $5.5B.
- Two-thirds of customer projects require an ISV or partner solution.

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AS/400 Channel and Selling Strategies

Selling Strategies
IBM currently positions the iSeries as having the following advantages, with Aberdeen's version of Sun's counter-positioning:

Table 8 — iSeries Selling Messages and Counters

iSeries Advantages by IBM	Draft Sun Advantages/Counterpoint
64-bit top to bottom for years	Ditto
99.94%+ single server reliability	Best data center Unix. Quote "nines".
Scalability with seamless growth	Best scalability story in the industry 15K
Hardware OS integration, OO architecture	That's the way IBM spells "proprietary!" SPARC and Solaris are well suited, but Solaris/Sun Linux on Intel makes Sun more open
Single-level storage	64-bit file system. The advantages of single-level storage have eroded over 15 years and are no longer a serious technology leadership factor
Dynamic LPAR	Solaris has been attacked by university student hackers for years. Solaris processes can be controlled a lot like OS/400 partitions. And Sun's partitioning is less likely to take down the whole system than IBM's share-a-processor approach.
Multiple NOS support	We support Linux too. [Sun: Insert Windows NT support here]
IXA adapter support for Windows NT consolidation	[Sun: Insert Windows NT support here]
Thousands of ISV solutions	Ditto.

Over 85% of iSeries machines are sold through the business partner channel. IBM direct sales reps get revenue credit for sales over $25,000 too, so everybody has some skin in the game. There is an elaborate lead referral program to the business partners, and extensive pre-sales support, benchmark centers, multi-partner coordinated selling, seminars, etc. We would hold up the iSeries business partner program against any in the industry for its overall success. However, the scope of this project does not include a detailed analysis of the business partner programs of IBM and Sun with an eye towards gap analysis or SWOT (strengths, weaknesses, opportunities, threats).

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iSeries Product Line

Common iSeries Features

Common to all of the models discussed below are:

- A sophisticated machine architecture that hides the hardware from the application and most OS functions.
- High-level language programming in C, Java, COBOL and legacy RPG;
- OS/400, a multi-user, multi-process OS with a long heritage of symmetric multiprocessing. OS/400 administration is considered particularly easy, and job control is sophisticated. Administration TCO is considered very low, and a point of customer admiration.
- A relational database, DB2/400, included with the OS. It is compatible with other implementations of DB2, but may not follow the same release cycle.
- An unusual single-level storage architecture that spreads data across all available disks, then takes advantage of the many disk arms to lower I/O contention. Maximizing CPU utilization on an iSeries always requires a room full of disk drives.
- Optional licensed support for "interactive workloads", IBM's terminology for legacy dumb-terminal applications. The license fees slide up with the number of simultaneous interactive users.
- Full, native support of IBM's middleware, especially WebSphere and Lotus Domino.
- Industry-standard desk-side and rack enclosures.

iSeries Models and Comparison

Table 9 depicts the five models of the iSeries product. Prices are U.S. list for a bootable but minimum system. The 250 is a uniprocessor, low-entry-price model used for development and low-volume distributed sites. The 270 is a one- or two-way system that is very popular with ISVs, and the volume leader in IBM's valued SMB market. The 2x0 machines compete with Sun's Netra line. The 8X0 machines are rack-mounted, data center symmetric multiprocessors which compete with Sun's midrange V880, 4800 and 6800. Aberdeen anticipates that the forthcoming iSeries 32-way models based on the POWER4 microprocessor will achieve 375,000 TPM-C — somewhat lower than the AIX version. This will move the iSeries competition up into the Sun Fire 12000 and 15000 family.

Table 9 — iSeries Models and Comparison to Sun

Model	Processors	Base $ (000s)	2001 Revenue ($Ms)	Sun Model Comparable
250	1	$9-14	$250	Ultra 10
270	1-2	$12-33	$3,500	220R
820	1-4	$64-267	$2,500	V880
830	1-8	$145-580	$2,018	4800
840	1-24	$860-1,555	$2,000	6800

Source Aberdeen Group

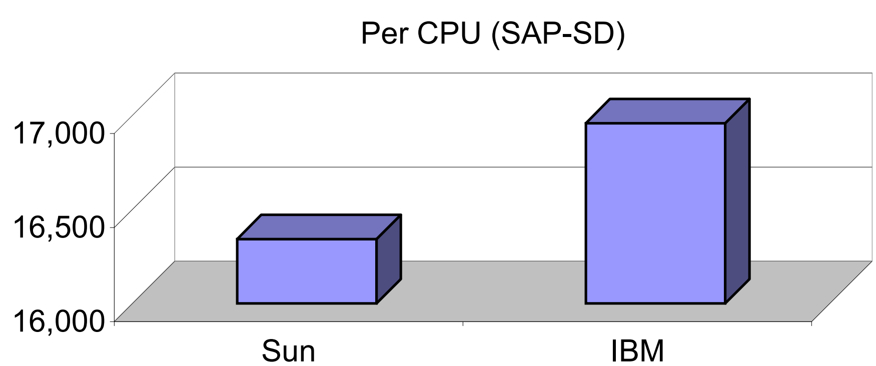
Performance and Price

It is impossible with public results to overlay the Sun and IBM iSeries product lines. Nevertheless, this section contains our analysis based on available information.

IBM has done workload characterization on all of its iSeries machines for years, and the company regularly publishes results in the form of a Commercial Processing Workload (CPW) for every model. Appendix B contains an analysis of the present iSeries product line with base pricing (CPU, base memory and cabinet, and minimum disk) as published at IBM's web site.

Based on the SAP SD benchmark of a fully loaded iSeries 840 and a fully loaded Star Fire 15000, each Star Fire CPU does 96% of the work of an iSeries 840 CPU, as illustrated in Figure 10.

Figure 10 — SAP SD Benchmark Results



Source: Aberdeen and SAP

There are no TPC benchmarks for comparison. However, a Sun relative, the Fujitsu Primepower provides a useful reference point, as depicted in Table 11.

Table 11 — TPC-C Benchmark Comparison

Product	CPUs	MHz	TPM-C	$/TPM-C	TPM-C/ CPU
Fujitsu Primepower 2000	128	563	455,818	$ 28.58	3,561
IBM 840-2420	24	450	163,776	$ 51.58	6,824

Source: Aberdeen and TPC

Based on the Notesbench Consortium's Notesmark, a fully configured V880 costing $176,000 does almost 20% of the work of a 24-way iSeries costing $3.2M. A 12-way Sun Fire 4800 costing $533,600 does 28% of the work of a 24-way iSeries costing $3.2M. (Note these comparisons are before Sun's April 2002 pricing action).

An analysis of SPEC JBB2000 shows Sun performing well, especially with the latest JVM, where the Sun Fire 6800 has twice the performance per CPU as the high-end iSeries 840-2420. By positioning with the 6800, Sun has competitive room to grow with the 12000 and 15000 models. See Table 12.

Table 12 — SPEC JBB2000 Benchmark Selected Results

Model	CPUs	SPEC JBB2000	JBB/CPU
IBM eServer iSeries 270-2253	2	7,500	3,750
IBM eServer iSeries 830-2402	4	17,864	4,466
IBM eServer iSeries 830-2403	8	34,274	4,284
IBM eServer iSeries 840-2420	24	80,348	3,348
Sun Enterprise-10000	64	230,049	3,595
Sun Fire 15K	72	324,309	4,504
Sun Fire 15K latest code	72	404,472	5,618
Sun Fire 6800	24	174,658	7,277

Source: Aberdeen and SPEC 4/9/02

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Opportunity Assessment

AS/400 Displacement Market Opportunity

Factors Positive to Sun Microsystems

7. *Technology Transition*: The iSeries is beginning a technology transition that will soon result in common hardware between IBM's Unix pSeries (e.g., p670 and p690 "Regatta") and Integrated Application iSeries. The positive aspects of this (see also the negative aspects below) are the increased competition between IBM business partners, caused by the lack of hardware differentiation between pSer ies (AIX Unix) and iSeries. This will cause IBM business partners to fight over operating systems differences in order to compete. Some of those business partners will want to switch to Sun in order to create a much greater line-in-the-sand differentiation. Note: our assumption is that hardware commonality will force IBM to narrow the price differences between the pSeries and the more expensive iSeries offerings.
8. *ISV Competition:* By marrying AIX and OS/400 in one chassis (see #1 above), IBM will unwittingly unleash two ISV camps that until this point were largely mutually exclusive; relatively few customers — and almost no SME customers — have both IBM Unix *and* AS/400 ISV applications. As a result, IBM ISVs will see new competition from other IBM business partners. This situation creates an opportunity for Sun to create fear, uncertainty, and doubt among *all* IBM business partners, while seriously courting a selected few.
9. *RAS*: The transition to the Regatta micro-architecture means iSeries customers must buy into IBM's new keep-running-or-fail philosophy — when a hard failure occurs, the Mean Time to Repair will be quite long compared to Sun due to the complexity of the backplane and multi-chip carrier design. We recommend that Sun begin a market education program about the availability implications of IBM's new design. Such a program will pay dividends competing against both the iSeries and the pSeries. (Aberdeen could help with this).
10. *Linux*: Many iSeries shops made a "no Unix" decision up to a decade ago and stuck with it. But the rapid emergence of Linux in the enterprise, heartily supported by IBM, creates a new opportunity for Sun. To succeed, Sun's Intel/Linux offering will have to be superior to IBM's iSeries-controlling-xSeries-Linux as well as IBM's native Linux-on-iSeries offering — on economic and technological levels as well as emotional. We conclude that Solaris-as-Linux could be attractive.
11. *Harmonious Trends*: Server consolidation is a marketing thrust of IBM, coinciding with an IT re-centralization trend, and continued interest in eBusiness application development leading to web services. We have seen server-per-country topologies condensed into two or three global data centers. The data centers

require large (profitable) configurations and enterprise levels of availability and serviceability. The downside is that consolidation typically involves some AS/400-specific applications in the initial phase, before a switch is made to eBusiness applications, implying the need to migrate these applications or keep them running.

12. *Sun has superior price and price-performance*. The just announced Oracle benchmarks with the 16-way p670 at 1.1 GHz are only 25% better than an 18-way 600 MHz p680 mid-range. This single data point suggests, on top of a lot of static microarchitecture analysis, that Regatta scaling will be an issue for some time to come — even as IBM transitions the iSeries to Regatta technology. While price-performance alone is an insufficient weapon to take on the AS/400 market, it doesn't hurt at all.

Factors Negative to Sun Microsystems

7. *More than 90% of the iSeries and AS/400 installed base is a combination of trailing edge, rather-die-than-switch mentality*. These operational managers grew up with the AS/400, have had ten years to evaluate Unix (and by implication Sun), and are not subject to a rational business discussion about the right platform supplier for their enterprise's future. And a work-around strategy to go directly to the CEO will show few results because the CEO believes IT is under control. Moreover, many of these Neanderthals are still happily pounding away at green-screen applications — with the implication that a total application migration must be part of any Sun attack strategy.

8. *IBM has done a good job of keeping the iSeries current in hardware and software* — successfully transitioning to RISC in the late 1990s and by offering partitioning, Lotus Domino, native Linux and Java, and management of xSeries servers. These technologies have allowed IBM's AS/400 to hold off Unix for a decade. Soon, IBM will announce that AS/400 customers can have it *both* ways, with a Regatta-class iSeries server able to run both OS/400 and AIX in different partitions. This will allow IBM customers to mix and match with best-of-breed Unix without giving up the benefits (and familiarity) of the iSeries. *We conclude that a technology-based market attack on the AS/400 would be relatively easy for IBM to deflect*.

9. *The AS/400 market is not static*. IBM allows district sales and country offices to experiment with new marketing and customer support programs as buyer interests and economic conditions change. If Sun is to meet the needs of today's iSeries buyers with Sun's SPARC servers and Solaris, it too should be prepared to anticipate buyer needs and meet them in the most profitable manner for Sun.

10. *IBM Global Services is a major recommender of ISV-based solutions sales.* They work with IBM business partners, and wouldn't think of recommending Sun.

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11. About *85%-90% of iSeries sales are made by IBM business partners* located in some 150 countries. IBM treats most ISVs as VARs, creating an economic incentive for the business partner to win with IBM. The iSeries is truly a global business for IBM, and Sun will have to be very aggressive to match IBM's coverage. Moreover, IBM has continued to sweeten the pot for business ISV's with incentives and marketing programs. The gross margins are high enough for the foreseeable future that IBM is unlikely to risk dissatisfying its business partners enough through declining incentives that ISVs will want to switch to Sun in large numbers voluntarily. We conclude that a strategy based on "buying ISVs" will cost too much.
12. *The lease renewal window which opened up for post-Y2K replacements is closing*. Three-year leases have already rolled. Five-year leases are up this year and next year. Presumably, IBM has already provided incentives for customers to roll over to another IBM lease. Obviously, it is hard to switch out a machine early in a lease due to financial penalties — and IBM is a formidable financing foe.

Sun's Addressable Market and Opportunity Size

Table 13 depicts in tabular form Aberdeen's analysis of the market opportunity and sizing for Sun based on flat IBM revenues for the iSeries in 2002 at $10.27 billion. Our conclusion is that the addressable market for Sun in 2002 is $1.87B.

Table 13 — 2002 Addressable Markets and Sizing

Addressable Market	Percent of HW/SW Revenues	Market Size $M	Yield Factor	Sun Addressable Market $M
New Customers	10%	$1,027	90%	$924
Consolidation	6%	$616	65%	$400
CRM, EAS, SCM, B2B	11%	$1,129	40%	$452
Upgrades	3%	$308	30%	$92
Total	30%	$3,080		$1,869

Source: Aberdeen Group

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- New Customers — represent 10% of iSeries revenues, $1,027M. We believe Sun can address 90% of this market, yielding an addressable market of $924M. In general, new customers are looking at an overall solution proposition, with the platform being only part of the solution. Sun plays well here. Negative yield factors include a relative lack of emerging market coverage and specialty ISV applications compared with IBM.
- Consolidation — represents 6% of IBM iSeries revenues. We believe 65% is addressable by Sun, yielding an addressable market of $400M. Negative yield factors are the need to migrate existing AS/400 applications, IBM financing, and IBM's proven consolidation services program.
- Applications — CRM, EAS, SCM, and B2B represent 11% of iSeries revenues. We believe that 40% is addressable, yielding an addressable market of $452M. Negative factors include IBM Global Services as recommender, existing relationships with ISVs and SIs as IBM business partners, and application add-ons to existing iSeries-specific ISV packages.
- Upgrades — Hardware and software upgrades related to Linux and xSeries management are estimated at 3% of revenues. This is the only portion of iSeries upgrades we find attractive. But selling Sun platforms as upgrades into iSeries shops is a low yield proposition, so we estimate only 30% is addressable — yielding $92M in addressable market for Sun.

Installed Base Migration is an Unattractive Market

iSeries customers leaving the IBM installed base are not covered in the addressable market above. First, we do not observe an identifiable class of dissatisfied iSeries users, so we cannot define an addressable market. Second, iSeries attrition to other platforms is already addressed by Unix and Windows NT competitors, so those sales are already being counted. Third, 75% of the installed base is in a low-growth/no-growth state of inertia. These customers are happy with their AS/400s and see little need to upgrade, let alone switch to Sun.

It is important for Sun management to understand that iSeries customers want IBM to make technology choices for them so that they can keep their MIS operations running. As a result, IBM has continued to charge a premium price for iSeries products while making long-term technology planning decisions for its iSeries customers.

The enterprises where AS/400s are most prevalent tend to look at IT as a cost of doing business, and therefore something to be tightly controlled and rationed. This group includes small businesses bounded by geography or specialty, quasi-regulated companies such as insurance firms and utilities, health care providers that are reimbursed by insurers, etc. Sun's Unix customers, as a class, tend to view IT as a lever to greater profits, and hence as a point of investment.

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AS/400 buyers are older, more senior-level personnel who are accustomed to mainframe prices and high support levels. To this class of customer, a "mainframe" class iSeries machine at under $1 million with an anticipated useful life of 5-7 years, with a quarter of the mainframe support costs, is highly attractive — even when these costs are twice what a Sun Solaris installation would be.

Aberdeen views a broad assault on the installed base as a low payback waste of resources. However, positioning Sun and Sun/Linux as a long-term leader in standard operating environments will yield capture sales over time as a normal part of general Sun marketing activities.

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Market Strategy

Aberdeen Group's recommended market strategy is to target selected segments of the iSeries market, as identified in the Market Opportunity section above.

New Customers
IBM wins new customers through its worldwide business partner organization. Sun has been and is already competing for these sales. To improve Sun's win ratio, we recommend:

- Focusing on the U.S., Europe, Japan and China as top geographies.
- Establishing a competitive hotline help desk for iSeries sales, staffed by iSeries-knowledgeable sales engineers.
- Educating Sun's direct sales and business partners with key counters to how IBM sells the iSeries and IBM's value proposition for the iSeries.
- Developing white papers and electronic collateral aimed specifically at the iSeries as new-customer competition.
- Identifying IBM's top ISVs, starting with Appendix B, and planning a Sun partner acquisition strategy unique to each top business partner and SI.
- Creating an iSeries-AS/400 competitive focus point in internal communications where wins and losses could be discussed and analyzed.
- Analyzing and countering IBM's finance and global services offerings related to new customers.

Consolidation
Consolidation often means centralizing small, distributed AS/400 system workloads back to a central data center. Sun is well positioned for enterprise-class, mission-critical commercial computing.

Identifying consolidation prospects can be done by analyzing commercial services (i.e. Gartner/Dataquest Computer Intelligence) for companies with multiple, distributed, aging AS/400 machines.

Only some consolidation opportunities are worth pursuing. Least attractive are custom, mission-critical applications written in AS/400 lock-in technology such as RPG. While there are some "liberation" toolsets from OS/400 to Solaris, the application migration issue will be the key bone of competition. More attractive are modern apps written in Java (but there are few of these) and cross-over ISVs such as Lawson who support Unix and OS/400 already.

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Application migration will require a defined services package with the ability to "fix price" the bulk of the services. This implies a specialized team of AS/400 conversion experts — or an outside partner with credibility. Accenture would be a candidate.

Sun must develop a life-cycle ROI model that factors in the migration costs to Solaris as well as the usual TCO factors. Sun's superior price-performance will be offset to some degree by the costs of migration and conversion services.

A typical consolidation sale should be a large 6800 or Sun Fire 12000-class workload in order to justify the pre-sales costs.

Applications — CRM, EAS, SCM, B2B

IBM, Sun, and Aberdeen all believe there is an attractive market for enterprise applications encompassing customer relationship management, ERP and enterprise applications, supply chain management, and business-to-business Internet infrastructure. IBM's present target is the small-medium business segment of this market. Aberdeen believes Sun should counter this program with its own.

Aberdeen Group has not been briefed on Sun's specific existing programs in the SMB application area. Therefore, we may be a little off target. That said, an application-driven marketing program would include the following elements:

- Orchestrating a global set of CRM, EAS, SCM and B2B ISVs, setting appropriate geographies and covering language and business practice differences.
- Recruiting and training a complementary set of systems integrators.
- Crafting appropriate levels of Sun services. Start with what IBM offers the SMB market for iSeries.
- Note that SMB enterprises largely do not require the rich functionality of a Siebel or SAP R3. Regional solutions may be highly salable because they are actually less functional but also less costly to train and therefore have a lower TCO.
- Sun ONE should be an adjunct to this effort. The thrust is to use Sun systems and best-of-breed applications instead of an application (and hardware and OS upgrade) to the installed AS/400. Focus on future-facing, investment protection, standards-based technology, lower cost of ownership. Intel-based Cobalt products with Sun Linux are an important part of the solution, as well as SPARC Solaris systems.

iSeries Upgrades — Linux is the Lever

In Table 13, we show the addressable market for upgrades as 3% of revenues, worth $308 million, whereas the total upgrades market is more than an order of magnitude greater. Sun cannot compete for the typical memory, CPU, or storage upgrades. Sun can compete for Linux and Linux appliance business, with a projected capture of $92 million.

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IBM supports Linux on the iSeries as a guest OS. But IBM has done a poor job of packaging Linux with appliance functionality.

We believe the overall enterprise level of interest in Linux, and the number of Linux-based applications, will skyrocket. Sun's market opportunity is to offer Linux to iSeries customers as an alternative to IBM's OS upgrade. The outline for such a program includes:

- Cobalt and Netra servers running general-purpose Linux applications, including Sun ISV applications that do not run on iSeries Linux.
- Cobalt appliances handling Internet Edge functions (i.e., H1).
- A mandatory technology component would be the ability to manage Sun servers from the OS/400 console, which we believe a web browser window would suffice and is already in place.
- A desirable technology component would be a Solaris/Linux shell which made Unix/Linux look like the OS/400 console commands and shell. Such a product may already be available, but we are unaware of it. The benefit is lower training costs and less aversion because the human interface would be familiar to change-averse iSeries technologists.
- Since IBM iSeries CPU cycles are expensive, a Netra/Cobalt program would "offer the price-performance of Linux while retaining the management "feel" of the AS/400".
- The program certainly applies to the F2000, but also can be sold as cost-saving-but-leading-edge technology to the SMB market.

Our vision is that Sun would treat Linux servers as non-threatening but complementary to the iSeries, just like many iSeries enterprise customers already share computing loads with Wintel servers managed by AS/400's and iSeries servers. We believe that Sun's Linux and appliance strategy and offerings can demand consideration at this time by iSeries customers when a Unix-based "SPARC Solaris is better" campaign would reach deaf ears.

Secondarily, the program would give Sun Linux business partners — including IBM business partners who want to hedge their Linux bets — the opportunity to sell both Linux business applications and appliances into the iSeries installed base.

Another opportunity, left unquantified, is opening the door to a discussion of a broader Sun-based relationship, probably with new ISV applications. This discussion plays with buyers as well as potential Sun ISVs who are considering switching.

Market Partner Strategies

Sun has to fight IBM's business partner "fire" with like fire: any Sun program that is not partner-driven will either fail or not reach the potential market. The key to the iSeries castle is through business partners.

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Systems Integrators
Sun already has business relationships with the major global systems integrators. Accenture, for example, has considerable experience with both iSeries and SPARC Solaris. Because the SIs are often called in to do the huge services portion of enterprise applications, they represent a short list for frank discussions. The SIs are a leverage point in the new-application market opportunity, where CRM, EAS, SCM, and B2B applications at the F2000 represent hundreds of millions in server revenues over the next several years.

ISVs and VARs
Application solutions have always been an important leg of the iSeries value proposition. Because of this, IBM has a well-oiled machine to attract, qualify, train, and support vertical market and horizontal ISVs in all geographies. One reason IBM keeps the prices of iSeries hardware and software high is to have the gross margin dollars to pay incentives to ISVs, drawing them closer to IBM and the iSeries base — hence the iNation initiative.

Any strategic move by Sun into the iSeries market would require a comprehensive comparative analysis of the IBM and Sun business partner programs, with a special emphasis on ISVs. In our opinion, IBM treats most ISVs like VARS, even if IBM gets the lead and does some of the selling, spreading enough money around to represent a serious portion of the ISV's marketing budget. Therefore, Sun needs to dissect IBM's business partner program in order to know the facts before it can develop a comprehensive counter program.

IBM's top-tier iSeries business partners are shown in Appendix B. A number of these (e.g., J.D. Edwards) are also members of iForce. Since these top business partners drive a large fraction of new iSeries sales, Sun should focus on these ISVs first. The smaller or more specialized ISVs are available by vertical market at IBM's business partner web site.

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Appendix A — Aggregation of Business Sizes

Statistics of U.S. Business: 1999
Management of companies & enterprises
United States

Employment size of enterprise	IBM Breakdown (small = 1-99, medium = 100-999, large=1000+ employees)									
	Number of companies				Number of employees			Annual payroll		
	# of companies	Sub-total	% Sub-total	% Sub-tot - ultra-small	Employees	Sub-total	% Sub-total	Payroll	Sub-total	% Sub-total
All firms	**5,607,743**				**110,705,661**			**$3,520,428,227**		
no employees (as of March 12)	709,074									
1 to 4 employees	2,680,087				5,606,302			143,112,925		
5 to 9 employees	1,012,954				6,652,370			166,598,812		
10 to 19 employees	605,693				8,129,615			217,571,005		
20 to 99 employees	501,848	5,509,656	98.25%	**98.00%**	19,703,162	40,091,449	**36.21%**	564,974,625	1,092,257,367	**31.03%**
100 to 499 employees	81,347				15,637,643			474,607,339		
500 to 999 employees	8,235	89,582	1.60%	**1.83%**	5,662,057	21,299,700	**19.24%**	181,765,746	656,373,085	**18.64%**
1000 to 2499 employees	4,992				7,629,355			254,215,167		
2500 to 4999 employees	1,706				5,904,452			211,574,407		
5000 to 9999 employees	871				6,064,760			225,756,578		
10000 or more employees	936	8,505	0.15%	**0.17%**	29,715,945	49,314,512	**44.55%**	1,080,251,623	1,771,797,775	**50.33%**

U.S. Department of Commerce latest release and Aberdeen Group

Appendix B — Top-Tier Business Partners

Company	Business Partner Level (Premier Level / Advanced Level)	Address	E-mail / URL
Alltel Information Services, Inc. IBM Developer Business Partner		2200 Lucien Way Suite 200 Maitland, Florida 32751 United States	
		Matching Solutions: Horizon Banking System	
DWL Incorporated IBM Developer Business Partner		230 Richmond St. E Level 2 Toronto, Ontario M5A 1P4 Canada	
		Matching Solutions: DWL Customer DWL Unifi	
e-Business Exchange Pte. Ltd. IBM Developer Business Partner		115, Jalan Chan Siew Teong Tanjung Bungah Georgetown 11200 Malaysia	
		Matching Solutions: e-BX BIll Presentment Switch Edition Version 3.0 e-BX Biller Service Provider Edition Version 3.0 e-BX TradePlus Merchant/Hosting Server Version 3.1.1	

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E3 Corporation IBM Developer Business Partner	1800 Parkway Place Suite 600 Marietta, Georgia 30067-7826 United States	
	Matching Solutions: E3 Marketplace.com E3CRISP E3SLIM E3TRIM	
EFA Software Services Ltd IBM Developer Business Partner	900, 311 6th Avenue SW. Calgary, Alberta T2P-3H2 Canada	
	Matching Solutions: Equator - Clearance & Settlement System Horizon - Automated Trading System	
Geac Enterprise Solutions IBM Developer Business Partner	Needles House Birmingham Road Studley, Warwickshire B80 7AS United Kingdom	
	Matching Solutions: Geac System21 - Service Management and Contract Modeller Geac System21 Customer Service and Logistics Applications Geac System21 Financial Applications Geac System21 for the Apparel, Footwear & Textile industries Geac System21 for the Automotive Industry Geac System21 for the Beverage Industry Geac System21 for the Food Industry	

Hyperion Solutions Corporation IBM Developer Business Partner	1344 Crossman Avenue Sunnyvale, California 94089-1113 United States **Matching Solutions:** Hyperion Performance Scorecard Hyperion Planning	
i2 Technologies Incorporated IBM Developer Business Partner	11701 Luna Rd. Dallas, Texas 75234 United States **Matching Solutions:** i2 Demand Planner Five.Two i2 Base Platform Five.Two i2 Demand Fulfillment Five.Two i2 Factory Planner Five.Two i2 High Availability Module Five.Two i2 Supply Chain Planner Five.Two	
IBS-US IBM Developer Business Partner	90 Blue Ravine Road Folsom, California 95630 United States **Matching Solutions:** IBS ASW Suite	

IBS AB IBM Developer Business Partner	Hemvärnsgatan 8 PO Box 1350 Solna 171 26 Sweden	
	Matching Solutions: ASW Alert Management ASW Analyser ASW Distribution ASW Financials ASW Manufacturing ASW NetStore / Internet Connection e2 ASW Sales and Marketing Support ASW Service IBS' Virtual Enterprise	
IBS Consist B.V. IBM Developer Business Partner	Nevelgaarde 20 P. O. Box 500 Nieuwegein 3430 AM Netherlands	
	Matching Solutions: ASW EMIS, Employee Management Information System FIS/2000, Financial Information System FMS, Financial Management System	
Infinium Software, Incorporated IBM Developer Business Partner	25 Communications Way 74 Furlong Rd Hyannis, Massachusetts 026012 United States	
	Matching Solutions: Infinium Business Intelligence Analytics (BIA) Infinium Financial Management Infinium Materials Management	

Intentia International AB IBM Developer Business Partner	Vendevägen 89 Box 596 Danderyd 182 15 Sweden	
	Matching Solutions: Movex Aviation Movex Distribution Movex e-CRM B2C Movex Fashion Movex Service & Rental Movex Steel	
International Business Systems N.V. IBM Developer Business Partner	Xavier de Cocklaan 70 St.-Martens-Latem B-9830 Belgium	
	Matching Solutions: AFS EE	
JD Edwards World Solutions Company IBM Developer Business Partner	One Technology Way Denver, Colorado 80237 United States	
	Matching Solutions: J.D. Edwards OneWorld Financial Management J.D. Edwards WorldSoftware®:	
JDA Software Group, Inc. IBM Developer Business Partner	14400 N 87th Street Scottsdale, Arizona 85260-3653 United States	
	Matching Solutions: E3TRIM and E3SLIM . Merchandise Management System (MMS®)	

Lawson Software IBM Developer Business Partner	380 St Peter Street St. Paul, Minnesota 55102 United States **Matching Solutions:** lawson.insight Procurement
Mincom Limited IBM Developer Business Partner	193 Turbot Street Brisbane, Queensland 4001 Australia **Matching Solutions:** Ellipse
Misys International Banking Systems IBM Developer Business Partner	45 Broadway New York, New York 10006 United States **Matching Solutions:** Equation DBA/Midas DBA
Misys International Banking Systems Limited IBM Developer Business Partner	1 St Georges Road Wimbledon London SW19 4DR United Kingdom **Matching Solutions:** Equation DBA Midas DBA

Qtech Business Systems Pty Ltd IBM Developer Business Partner	Level 1 16 Edmondstone Street Brisbane, Queensland 4051 Australia	
	Matching Solutions: Guardianship	
SAP AG IBM Developer Business Partner	Neurottstr. 16 Walldorf D-69190 Germany	
	Matching Solutions: mySAP E-Procurement mySAP Financials mySAP Human Resources mySAP Marketplace mySAP Product Lifecycle Management mySAP Workplace	
SAP France IBM Developer Business Partner	57 59 BLD Malsherbes Paris 75008 France	
	Matching Solutions: mySAP.com Business Warehouse	
Sapiens International IBM Developer Business Partner	2000 CentreGreen Way Suite 240 Cary, North Carolina 27513 United States	
	Matching Solutions: bizBLOX	

SAS Institute Inc. IBM Developer Business Partner	SAS Campus Drive Cary, North Carolina 27513 United States	
	Matching Solutions: SAS/IntrNet (R) Software	
Siebel Systems, Inc. IBM Developer Business Partner	1855 South Grant Street San Mateo, California 94402-2667 United States	
	Matching Solutions: Siebel 7, MidMarket Edition Siebel Call Center Siebel eBusiness Solutions Siebel Enterprise Applications Siebel Field Service Siebel Sales 2000	
Silvon Software, Inc. IBM Developer Business Partner	900 Oakmont Lane Suite 400 Westmont, Illinois 60559 United States	
	Matching Solutions: Stratum	
Software AG (Germany) IBM Developer Business Partner	Uhlandstr. 12 Darmstadt 64297 Germany	
	Matching Solutions: EntireX	

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SOPRA GROUP IBM Developer Business Partner	9 Bis rue de Presbourg Paris 75116 France	
	Matching Solutions: XFB eXtended File Broker	
Vision Solutions, Inc. IBM Developer Business Partner	17911 Von Karman Avenue 5th Floor Irvine, California 92614 United States	
	Matching Solutions:	

Appendix C — iSeries Models, Prices & Performance

iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
SB2_2315	$282,890	8	7,350		70	4	72,955	181,300	50,560
250_0297	$8,500	1	50	15	175	4	496	1,233	344
250_0298	$12,000	1	75	20	175	4	744		
250_2295	$8,700	1	50	15	175	10	496		
250_2296	$12,300	1	75	20	175	10	744		
270_2248_1517	$12,088	1	150	25	843	24	1,489	3,700	1,032
270_2250_1516	$13,288	1	370	0	843	24	3,673		
_________1518	$32,788			30					
270_2252_1516	$25,288	1	950	0	843	24	9,430	23,433	6,535
_________1519	$81,588			50					
270_2253_1516	$48,288	2	2,000	0	843	24	19,852	49,333	13,758
_________1520	$160,788			70					
270_2422	$11,288	1	50	0	843	24	496	1,233	344
270_2423	$15,288	1	100	0	843	24	993		688
270_2424	$22,288	2	200	0	843	24	1,985		
270_2431_1518	$32,276	1	465	30	843	24	4,616		
270_2432_1516	$15,476	1	1,070	0	843	24	10,621	26,393	7,360
_________1519	$71,776			50					
270_2434_1516	$31,788	2	2,350	0	843	24	23,326	57,967	16,165
_________1520	$144,288			70					
270_2452	$13,776	1	100		843	24	993	2,467	688
270_2454	$24,788	2	240		843	24	2,382		
820_0150	$63,900	1	1,100		8,334	237	10,918		
820_0151	$97,400	2	2,350		8,334	237	23,326		
820_0152	$139,900	4	3,700		8,334	237	36,726	91,267	25,452

iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
820_2395_1521	$39,400	1	370	35	8,334	237	3,673	9,127	2,545
_________1522	$67,400			70					
_________1523	$119,400			120					
_________1524	$194,400			240					
820_2396_1521	$96,900	1	950	35	8,334	237	9,430	23,433	6,535
_________1522	$124,900			70					
_________1523	$176,900			120					
_________1524	$251,900			240					
_________1525	$411,900			560					
820_2397_1521	$174,900	2	2,000	35	v 8334	237	19,852	49,333	13,758
_________1522	$202,900			70					
_________1523	$254,900			120					
_________1524	$329,900			240					
_________1525	$489,900			560					
_________1526	$704,900			1,050					
820_2398_1521	$266,900	4	3,200	35	v 8334	237	31,763	78,933	22,012
_________1522	$294,900			70					
_________1523	$346,900			120					
_________1524	$421,900			240					
_________1525	$581,900			560					
_________1526	$796,900			1,050					
_________1527	$1,091,900			2,000					
820_2425	$31,900	1	100	-	8,334	237	993	2,467	688
820_2426	$38,900	2	200	-	8,334	237	1,985	4,933	1,376
820_2427	$45,900	4	400	-	8,334	237	3,970	9,867	2,752
820_2435_1521	$60,900	1	600	35	8,334	237	5,955	14,800	4,127
_________1522	$88,900			70					
_________1523	$140,900			120					

iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
_________1524	$215,900			240					
820_2436_1521	$96,900	1	1,100	35	8,334	237	10,918	27,133	7,567
_________1522	$124,900			70					
_________1523	$176,900			120					
_________1524	$251,900			240					
_________1525	$411,900			560					
820_2437_1521	$174,900	2	2,350	35	8,334	237	23,326	57,967	16,165
_________1522	$202,900			70					
_________1523	$254,900			120					
_________1524	$329,900			240					
_________1525	$489,900			560					
_________1526	$704,900			1,050					
820_2438_1521	$259,900	4	3,700	35	8,334	237	36,726	91,267	25,452
_________1522	$287,900			70					
_________1523	$339,900			120					
_________1524	$414,900			240					
_________1525	$574,900			560					
_________1526	$789,900			1,050					
_________1527	$1,084,900			2,000					
820_2456	$30,400	1	120	0	8,334	237	1,191	2,960	825
820_2457	$41,400	2	240	0	8,334	237	2,382		
820_2458	$49,900	4	380	0	8,334	237	3,772		
830_2351_1531	$145,330		1,200	70	22,153	630	11,911	29,600	8,255
_________1532	$260,330			120					
_________1533	$335,330			240					
_________1534	$495,330			560					
_________1535	$710,330			1,050					
_________1536	$1,005,330			2,000					

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iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
_________1537	$1,505,330		7,350	4,550			72,955	181,300	50,560
830_2400_1531	$170,130	2	1,850	70	22,153	630	18,363		
_________1532	$285,130			120				0	
_________1533	$360,130			240					
_________1534	$520,130			560					
_________1535	$735,130			1,050					
830_2402_1531	$355,330	4	4,200	70	22,153	630	41,688	103,600	28,891
_________1532	$470,330			120					
_________1533	$545,330			240					
_________1534	$705,330			560					
_________1535	$920,330			1,050					
_________1536	$1,215,330			2,000					
830_2403_1531	$580,330	8	7,350	70	22,153	630	72,955	181,300	50,560
_________1532	$695,330			120					
_________1533	$770,330			240					
_________1534	$930,330			560					
_________1535	$1,145,330			1,050					
_________1536	$1,440,330			2,000					
_________1537	$1,940,330			4,550					
840_2352_1540	$862,268	37480	9,000	120	37,978	1080	89,332	222,000	61,910
_________1541	$1,052,268			240					
_________1542	$1,212,268			560					
_________1543	$1,427,268			1,050					
_________1544	$1,722,268			2,000					
_________1545	$2,222,268			4,550					
_________1546	$3,292,268		12,000	10,000					
840_2353_1540	$1,112,268	37608	12,000	120	37,978	1080	119,110	296,000	82,546
_________1541	$1,302,268			240					

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iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
_________1542	$1,462,268			560					
_________1543	$1,677,268			1,050					
_________1544	$1,972,268			2,000					
_________1545	$2,472,268			4,550					
_________1546	$3,542,268			10,000					
_________1547	$4,812,268		16,500	16,500			163,776	407,000	113,501
840_2354_1540	$1,482,268	18 / 24	16,500	120	37,978	1080	163,776	407,000	113,501
_________1541	$1,672,268			240					
_________1542	$1,832,268			560					
_________1543	$2,047,268			1,050					
_________1544	$2,342,268			2,000					
_________1545	$2,842,268			4,550					
_________1546	$3,912,268			10,000					
_________1547	$5,182,268			16,500					
_________1548	$5,807,268			20,200					
840_2416_1540	$722,268	37480	7,800	120	37,978	1080	77,421	192,400	53,655
_________1541	$912,268			240					
_________1542	$1,072,268			560					
_________1543	$1,287,268			1,050					
_________1544	$1,582,268			2,000					
_________1545	$2,082,268			4,550					
_________1546	$3,152,268		10,000	10,000			99,258	246,667	68,789
840_2417_1540	$887,268	37608	10,000	120	37,978	1080	99,258	246,667	68,789
_________1541	$1,077,268			240					
_________1542	$1,237,268			560					
_________1543	$1,452,268			1,050					
_________1544	$1,747,268			2,000					
_________1545	$2,247,268			4,550					

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iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
_________1546	$3,317,268		13,200	10,000			131,021	325,600	90,801
840_2418_1540	$782,268	12	10,000	120	37,978	1080	99,258	246,667	68,789
_________1541	$972,268			240					
_________1542	$1,132,268			560					
_________1543	$1,347,268			1,050					
_________1544	$1,642,268			2,000					
_________1545	$2,142,268			4,550					
_________1546	$3,212,268			10,000					
840_2419_1540	$1,132,268	18 / 24	13,200	120	37,978	1080	131,021	325,600	90,801
_________1541	$1,322,268			240					
_________1542	$1,482,268			560					
_________1543	$1,697,268			1,050					
_________1544	$1,992,268			2,000					
_________1545	$2,492,268			4,550					
_________1546	$3,562,268			10,000					
_________1547	$4,832,268		16,500	16,500			163,776		
840_2420_1540	$1,272,268	24	16,500	120	37,978	1080	163,776	407,000	113,501
_________1541	$1,462,268			240					
_________1542	$1,622,268			560					
_________1543	$1,837,268			1,050		9.9258			
_________1544	$2,132,268			2,000					
_________1545	$2,632,268			4,550					
_________1546	$3,702,268			10,000					
_________1547	$4,972,268			16,500					
840_2461_1540	$1,555,268	24	20,200	120	37,978	1080			138,953
_________1541	$1,745,268			240					
_________1542	$1,905,268			560					
_________1543	$2,120,268			1,050					

iSeries System Model	Price (US$) Minimum Config.	n-way CPUs (max.)	Pro-cessor CPW	Inter-active CPW	Disk GB	Disk Arms	TPM-C Estimate	SAP SD Estimate	Notes Bench Estimate
_________1544	$2,415,268			2,000					
_________1545	$2,915,268			4,550					
_________1546	$3,985,268			10,000					
_________1547	$5,255,268			16,500					
_________1548	$5,880,268			20,200					

Notes:

Prices are as of 7 April 2002 From www.ibm.com

$ Price (in black) represents a packaged offering which includes software and more hardware than a minimally operational machine.

$ Price represents a minimally operational machine which includes the minimum hardware to successfully perform an initial program load.

9.925818182 ratio of TPM-C to Processor CPW based on 840_2420 3/19/2001 (Source: Aberdeen Group)

24.66666 ratio of SAP SD to Processor CPW on 840_2420 6/21/01 (Source: Aberdeen Group)

6.878861386 ratio of Notes Bench to Processor CPW on 840_2420 6/21/01 (Source: Aberdeen Group)

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